Architecture	CIFAR-10	CIFAR-100
DenseNet-BC	5.0	19.9
WRN-28-10	5.0	20.4

Table 1: Test error rates on CIFAR-10 and CIFAR-100

of the algorithms are described and conclusions are drawn. Finally, our method is compared against the current state-of-the-art ODIN [13] and is shown to significantly outperform it.

4.1 Experimental Setup

We use CIFAR-10 (contains 10 classes) and CIFAR-100 (contains 100 classes) [10] datasets as in-distribution datasets to train deep neural networks for image classification. They both consist of 50,000 images for training, and 10,000 images for testing. The dimensions of an image in both the datasets is 32×32. The classes of both CIFAR-10 and CIFAR-100 are randomly divided into five parts. As described in Section 3.2, each part is assigned as OOD to a unique network which is then trained. For each network, the other parts act as in-distribution samples.

Following the benchmarks given in [13], the following OOD datasets are used in our experiments. The datasets are described in ODIN[13] and provided as a part of their code release; here we are simply restating the description for comprehensiveness.

- **TinyImageNet[9] (TIN)** is a subset of ImageNet dataset[13]. Tiny ImageNet contains 200 classes which is drawn from original 1,000 classes of ImageNet. In total, there are 10,000 images in the Tiny ImageNet. By randomly cropping and downsampling each image to 32×32, two datasets *TinyImageNetcrop* (*TINc*) and *TinyImageNetresize* (*TINr*) are constructed.
- **LSUN** is the Large Scale UNderstanding dataset (LSUN)[21] created by Princeton, using deep learning classifiers with humans in the loop. It contains 10,000 images of 10 scene categories. By randomly cropping and downsampling each image to size 32×32, two datasets *LSUNc* and *LSUNr* are constructed.
- **iSUN[20]** is collected by gaze tracking from Amazon Mechanical Turk using a webcam. It contains 8925 scene images. Similar to the above dataset as other datasets, images are down-sampled to size 32×32.
- **Uniform Noise (UNFM)** is synthetic dataset consists of 10,000 noise images. The RGB value of each pixel in an image is drawn from uniform distribution in the range $[0, 1]$.
- **Gaussian Noise (GSSN)** is synthetic dataset consists of 10,000 noise images. The RGB value of each pixel is drawn from independent and identically distributed Gaussian with mean 0.5 and unit variance and each pixel value is clipped to the range $[0, 1]$.

Neural network architecture Following ODIN[13], two state-of-the-art neural network architectures, *DenseNet* [8] and *Wide ResNet*(WRN) [22], are adopted to evaluate our method. For DenseNet, we use the DenseNet-BC setup as in [8], with depth $L = 100$, growth rate $k = 12$ and dropout rate 0. For Wide ResNet, we use WRN-28-10 setup, with depth 28, width 10 and dropout rate of 0.3. We train both DenseNet-BC and Wide ResNet on CIFAR-10 and CIFAR-100 for 100 epochs with batch size 100, momentum 0.9, weight decay 0.0005, and margin 0.4. The initial training rate is 0.1 and it is linearly dropped to 0.0001 over the whole training process. During training, we augment our training data with random flip and random cropping. We use the smallest OOD dataset, iSUN, as validation data for hyper-parameter search. We test the rest four out-of-distribution datasets except iSUN on our trained network. During testing, we use batch size 100. Similar to ODIN[13], input preprocessing with $\epsilon = 0.002$ is used.

Table 1 shows the test error rates when our method is trained and tested on CIFAR-10 and CIFAR-100 respectively using the algorithms 1 and 2. For CIFAR-10, the vanilla DenseNet-BC [8] and the proposed method gives error rates of 4.51% and 5.0% respectively. For CIFAR-100, the error rates are 22.27% and 19.9% respectively. On both these datasets, the difference in error rates is marginal. For WRN [22] with depth 40, $k = 10$, the test error rate on CIFAR-10 for the vanilla network is 4.17% and for the proposed network is 5.0%. For CIFAR-100, the error rates are 20.5% and 20.4% for the vanilla network and the proposed network respectively. The small difference in the performance on CIFAR-10 can be explained by the fact that the our method did not use the ZCA whitening preprocessing while the vanilla network did.

4.2 Evaluation Metrics

To measure the effectiveness of our method to distinguish between in-distribution and out-of-distribution samples, we adopt five different metrics, same as what was used in ODIN[13] paper. We restate these metrics below for comprehensiveness. In the rest of manuscript, TP, TN, FP, FN are used to denote true positives, true negatives, false positives and false negatives respectively.

FPR at 95% TPR measures the probability that an out-of-distribution sample is misclassified as in-distribution when the true positive rate (TPR) is 95%. In this metric, TPR is computed by $TP/(TP+TN)$, and FPR is computed by $FP/(FP+TN)$.

Detection Error measures the minimum misclassification probability over all possible score thresholds, as defined in ODIN[13]. To have a fair comparison with ODIN, the same number of positive and negative samples are used during testing.

AUROC is the Area Under the Receiver Operating Characteristic curve. In a ROC curve, the TPR is plotted as a function of FPR for different threshold settings. AUROC equals to the probability that a classifier will rank a randomly chosen positive sample higher than a randomly chosen negative one. AUROC score of 100% means perfect separation between positive and negative samples.

Ablation Studies	Parameters	FPR at 95% TPR ↓	Detection Error ↓	AUROC ↑	AUPR In ↑	AUPR Out ↑	CLS Acc ↑
Number of Splits	3	32.37	13.94	93.50	94.39	92.22	76.41
	5	**22.95**	**10.79**	**95.69**	**96.55**	94.3	80.01
	10	28.71	12.53	94.48	95.37	93.26	81.94
	20	23.85	10.95	95.49	96.24	**94.36**	**82.33**
Type of splits	Random	**22.95**	**10.79**	**95.69**	**96.55**	**94.3**	**80.01**
	Manual	40.16	16.26	91.57	92.90	89.57	79.79
Epsilon	0.000000	53.51	16.37	90.75	93.16	86.71	**80.32**
	0.000313	41.62	14.37	92.8	94.52	90.16	80.22
	0.000625	34.64	12.83	94.09	95.4	92.19	80.17
	0.001250	25.74	11.19	95.38	96.31	94.04	80.08
	0.002000	**22.95**	**10.79**	**95.69**	**96.55**	**94.3**	80.01
	0.003000	29.07	11.79	94.73	95.9	92.43	79.97
Temp-rature	1	38.57	17.32	91.44	92.7	90.12	**80.01**
	10	27.84	11.93	94.86	95.81	93.39	**80.01**
	100	24.44	10.86	95.6	96.5	94.17	**80.01**
	1000	22.95	**10.79**	**95.69**	**96.55**	**94.3**	**80.01**
	5000	**22.7**	10.81	95.66	96.53	94.28	**80.01**
Loss Function	SFX	84.09	36.55	68.96	72.38	63.77	54.18
	SFX+MaxEntropyDiff	50.70	19.65	88.26	89.71	86.18	72.99
	SFX+MarginEntropy	**22.95**	**10.79**	**95.69**	**96.55**	**94.3**	**80.01**
OOD Detection Score	SFX	50.52	19.91	88.69	90.91	86.19	**80.01**
	Entropy	36.23	16.48	91.92	93.03	90.74	**80.01**
	SFX+Entropy	38.57	17.32	91.44	92.7	90.12	**80.01**
	SFX@Temp	**22.71**	10.83	95.65	96.52	94.26	**80.01**
	Entropy@Temp	37.0	14.05	93.33	94.76	91.04	**80.01**
	(SFX+Entropy)@Temp	22.95	**10.79**	**95.69**	**96.55**	**94.3**	**80.01**

Table 2: Ablation Studies on CIFAR-100 as in-distribution data and iSUN as out-of-distribution data on DenseNet-100 network. All values are percentages. ↑ indicates larger value is better, and ↓ indicates lower value is better.

AUPR-In measures the Area Under the Precision-Recall curve. In a PR curve, the *precision*= $TP/(TP+FP)$, is plotted as a function of *recall*= $TP/(TP+FN)$, for different threshold settings. Since precision is directly influenced by class imbalance (due to FP), PR curves can highlight performance differences that are lost in ROC curves for imbalanced datasets[5]. AUPR score of 100% means perfect distinguish between positive and negative samples. For AUPR-In metric, in-distribution images are specified as positive.

AUPR-Out is similar to the metric AUPR-In. The difference lies in that for AUPR-Out metric, out-of-distribution images are specified as positive.

CLS Acc is the classification accuracy for ID samples.